Matrix Diagonalization

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Introduction to Diagonalization

Definition

- Let $A, B \in M_n(\mathbb{K})$, we say that A and B are **similar** if there exists an invertible matrix $P \in M_n(\mathbb{K})$ such that
$$\boxed{B = P^{-1}AP}$$
- We say that the matrix A is **diagonalizable** when there exists a diagonal matrix $D \in M_n(\mathbb{K})$ such that A and D are similar.

Example :

1) Show that the matrices $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{pmatrix}$ are similar using the matrix $P = \begin{pmatrix} \frac{-1+\sqrt{5}}{2} & \frac{-1-\sqrt{5}}{2} \\ 1 & 1 \end{pmatrix}$.

2) Deduce that A is diagonalizable.

Techniques to determine the similarity of matrices

a) Using the determinant :

Proposition

If A and B are similar, then

$$\boxed{det(A) = det(B)}$$

Example :

Show that the matrices $A = \begin{pmatrix} 2 & (-3) \\ 4 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ are not similar.

b) Using the trace :

Proposition :

Let $A, B \in M_n(\mathbb{R})$, then we have

1. $\mathrm{tr}(A \times B) = \mathrm{tr}(B \times A)$.
2. If A and B are similar, then $\mathrm{tr}(A) = \mathrm{tr}(B)$

Example :

Show that the matrices $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & (-1) \\ 1 & 0 \end{pmatrix}$ are not diagonalizable.

Diagonalization and linear applications

Proposition

Let E be a Vector Space of finite dimension, $f : E \to E$ a linear application and $\mathcal{B}, \mathcal{B}'$ two bases of E. If we denote

$$A = \mathrm{Mat}_{\mathcal{B},\mathcal{B}}(f) \text{ and } B = \mathrm{Mat}_{\mathcal{B}',\mathcal{B}'}(f),$$

then :

1. A and B are similar and we have
$$\boxed{B = T^{-1}AT}$$
where T is the transition matrix $T_{\mathcal{B},\mathcal{B}'}$.
2. If C is a matrix of the same size as A such A and C are similar, then there exists a basis $\mathcal{B}''$ such that
$$\boxed{C = \mathrm{Mat}_{\mathcal{B}'',\mathcal{B}''}(f)}$$

Proposition

Let $A \in M_n(\mathbb{K})$ and let $f : \mathbb{K}^n \to \mathbb{K}^n$ be the linear application defined by A, i.e. $\forall v \in \mathbb{K}^n, f(v) = A \times v$. Then
A is diagonalizable $\Leftrightarrow$ there exists a basis $\{e_1, \dots, e_n\}$ of $\mathbb{K}^n$ such that

$$\forall i \in \{1, \dots, n\}, \exists \lambda_i \in \mathbb{K}, f(e_i) = \lambda_i e_i.$$

In which case, if we let P the matrix whose columns are the vectors e_i, then we have

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$

Consequence

Let $f \in \mathcal{L}(E)$ where E is a Vector Space of finite dimension. We say that f is diagonalizable if there exists a basis $\mathcal{B}$ of E composed of eigen-vectors of f such that $\mathrm{Mat}_{\mathcal{B}}(f)$ is diagonal.

Definition

- Let $f : E \to E$ be a linear application. We call an **eigen-vector** of f every vector $v \neq 0$ such that $f(v) = \lambda \cdot v$ with $\lambda \in \mathbb{K}$.
- The scalar λ such that $f(v) = \lambda \cdot v$ is called an **eigen-value** of f.
- We say that the vector v is associated to the eigen-value value λ.
- For a fixed $\lambda \in \mathbb{K}$, the set of vectors $v \in E$ such that $f(v) = \lambda \cdot v$ is called the associated **eigen-space** to λ and is denoted E_λ.

Example :

Let the matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Find the matrix P such that

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

Proposition

- Let $A \in M_n(\mathbb{K})$ and let f be the application such that $f(v) = Av$. Then

$$\boxed{\lambda \text{ is an eigen-value of } f \Leftrightarrow \det(A - \lambda I_n) = 0}$$

- The expression $\det(A - \lambda I_n)$ is a polynomial of the variable λ called the **characteristic polynomial** of A (or of f) and denoted χ_A (or χ_f).

Example :

1) Let the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Find the characteristic polynomial of A.

2) Let the matrix $A = \begin{pmatrix} 6 & 2 & 1 \\ (-6) & (-1) & (-2) \\ 0 & 0 & 3 \end{pmatrix}$.

a) Find the characteristic polynomial of A,
b) Find the eigen-values and the eigen-vectors of A,
c) Diagonalize A.

Proposition

Let A and B two matrices of $M_n(\mathbb{K})$. If A and B are similar, then $\chi_A = \chi_B$.

Proposition

Let $f : E \to E$ be a linear application, and let $\{e_1, \ldots, e_n\}$ be a set of eigen-vectors of f.
We suppose furthermore that $\forall i \in \{1 \ldots n\}$, the vector e_i is associated to the eigen-value λ_i.
$\Rightarrow$ If the scalars $\lambda_1, \ldots, \lambda_n$ are distinct, then the set $\{e_1, \ldots, e_n\}$ is linearly independent

Proposition

Let $A \in M_n(\mathbb{K})$ and let f be the associated linear application to A. Let $\lambda_1, \lambda_2, \ldots, \lambda_m$ be the distinct roots of the characteristic polynomial of A. For each λ_i, let n_i be the dimension of the associated eigen-space $E_{\lambda_i} = Ker(f - \lambda_i Id_E)$, and let $\{e_{i1}, e_{i2}, \ldots, e_{in_i}\}$ be a basis of E_{λ_i}.
Then

$$\boxed{A \text{ is diagonalizable if and only if } \sum_{i=1}^{m} n_i = n}$$

In which case, the set $\{e_{ij}\}_{i=1\ldots m, j=1\ldots n_i}$ is a basis of $\mathbb{K}^n$ composed of eigen-vectors of f.

Consequence

Let $A \in M_n(\mathbb{K})$. If the characteristic polynomial of A has n distinct roots of $\mathbb{K}$, then A is diagonalizable.

Exercise :

Show that the following matrix is diagonalizable :

$$A = \begin{pmatrix} 3 & 6 & (-8) \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix}$$

Theorem

Let $A \in M_n(\mathbb{K})$. Then A is diagonalizable if and only if

1. its characteristic polynomial χ_A is written as
$$\boxed{\chi_A(\lambda) = (-1)^n(\lambda - \lambda_1)^{\alpha_1}(\lambda - \lambda_2)^{\alpha_2} \dots (\lambda - \lambda_p)^{\alpha_p}}$$
with $\lambda_1, \dots, \lambda_p \in \mathbb{K}$ and $\alpha_1 + \dots + \alpha_p = n$.
$\Rightarrow$ α_i is called the **multiplicity** of the eigen-value λ_i.
2. the dimensions of the eigen-spaces E_{λ_i} are maximal, i.e.
$$\boxed{\dim(E_{\lambda_i}) = \alpha_i}$$

Exercise :

Show that the following matrix is diagonalizable :

$$A = \begin{pmatrix} (-1) & 1 & 1 \\ 1 & (-1) & 1 \\ 1 & 1 & (-1) \end{pmatrix}$$

END

References :

-*Pierre Cuillot, Cours Concis, IRMA*.